



Homework 2

Deadline: 2025/11/25 23:59:59

[Kaggle Link](#)



UAV Segmentation

Semantic Segmentation:

- Unlike classification: predicts a label for each pixel
- Enables deeper scene understanding

Competition Task

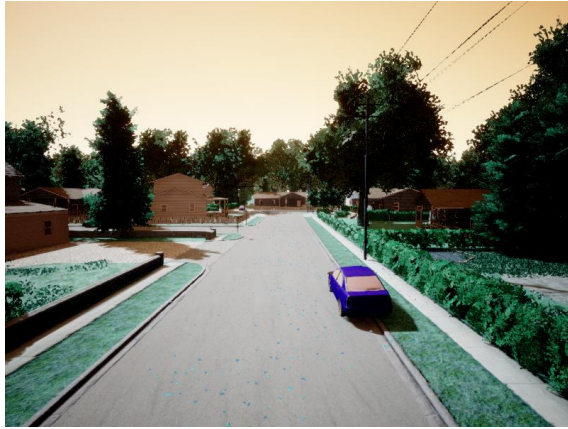
- Dataset: synthetic UAV images generated with **Unreal Engine**
- Each image paired with a **grayscale segmentation mask**
- Mask pixel values: 0–15 $\rightarrow$ 16 semantic classes
- Goal: train a model to generate accurate masks for unseen images



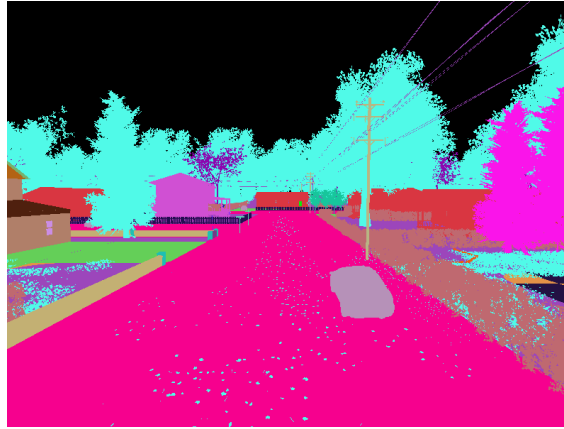


UAV Segmentation

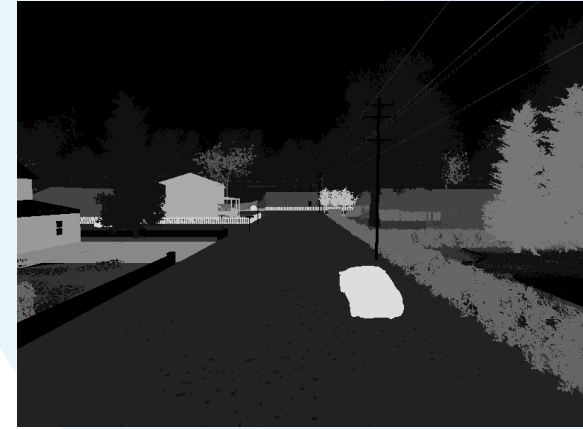
The dataset consists of UAV aerial images annotated with **16 semantic classes**, including trees, roads, houses, vehicles, sky, and other common objects, providing a comprehensive representation of real-world outdoor scenes.



Origin



RGB



Gray-scale



UAV Segmentation

UAV_dataset:

- Train: 4000 images, 4000 masks - Weather: Normal/Rain/Snow/Fallen
- Test: 1000 images - Weather: Normal/Rain/Snow/Fallen/Dust/Fog

UAV_dataset (2 directories)

About this directory

This directory does not have a description yet.

test
1 directories

train
2 directories

Data Explorer

3.78 GB

UAV_dataset

Summary

9000 files

[Download All](#)

[New Version](#)

DOWNLOAD DATA

```
>_ kaggle competitions download -c 2025-ncku-ee-ml-16-classes-segmentation
```

Link : <https://www.kaggle.com/competitions/2025-ncku-ee-ml-16-classes-segmentation/data>



Kaggle competition

KANESADOA · COMMUNITY PREDICTION COMPETITION · PRIVATE · 2 MONTHS TO GO

2025 NCKU EE ML UAV Segmentation

Develop segmentation models to classify 16 categories from UAV aerial images simulated with Unreal Engine.

[Join Competition](#)

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Overview

Welcome to the UAV Semantic Segmentation Challenge

In this competition, your task is to develop a semantic segmentation model that can accurately classify aerial scenes into 16 categories. The dataset has been generated using Unreal Engine, simulating UAV (drone) images of urban and natural environments.

Start
 18 seconds ago

Close
 2 months to go

Competition Host

kanesadoa

Prizes & Awards

Kudos
Does not award Points or Medals

Participation

0 Entrants
0 Participants
0 Teams
0 Submissions

Tags

Custom Metric

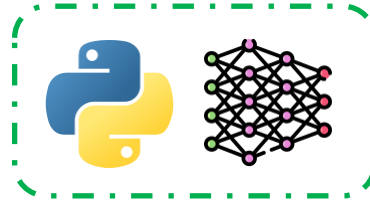


Kaggle competition

Input folder



4000.png
4001.png
.....
4999.png



python program



submission.csv

2025 NCKU EE ML UAV Segmentation

Overview Data Code Models Discussion Leaderboard Rules Team Submissions Se

Submission Code

Here is an example code snippet to help you generate a submission file in the correct format.

```
import os
import cv2
import pandas as pd
import numpy as np

def rle_encode(mask):
    pixels = mask.flatten(order='F')
    pixels = np.concatenate([[0], pixels, [0]]) # padding
    runs = np.where(pixels[1:] != pixels[:-1])[0] + 1
    runs[1::2] -= runs[:-1:2]
    return ' '.join(str(x) for x in runs)

# 資料夾和輸出的csv存放路徑
mask_dir = './test_data'
output_csv = 'sample_submission.csv'
num_classes = 16 # 0~15 類別
```

Convert folder's gray-scale
images to CSV format



Kaggle competition

Search

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2025 NCKU EE ML UAV Segmentation

Develop segmentation models to classify 16 categories from UAV aerial images simulated

Overview Data Code Models Discussion Leaderboard Rules Team

Overview

Welcome to the UAV Semantic Segmentation Challenge

In this competition, your task is to develop a semantic segmentation model that can accurately classify 16 categories. The dataset has been generated using Unreal Engine, simulating UAV (drone) images of environments.

Start
3 hours ago

Description

Semantic segmentation is one of the core challenges in computer vision. Unlike classification, where a single label to an image, segmentation requires predicting a class label for every pixel in the image. To gain a deeper understanding of scene structure, which is essential for applications such as autonomous driving, we have generated a dataset of simulated UAV images and to analyze traffic.

Submit to Competition

File Upload Notebook

2025 NCKU EE ML UAV Segmentation

You have 20 submissions remaining today. This resets in 12 hours.

Drag and drop file to upload
(e.g., .csv, .parquet, .zip, .gz, .7z, .tar)

or

Browse Files

Your submission should be a CSV or Parquet file with 1000 rows and a header. You can upload a zip/gz/7z/tar archive.

SUBMISSION DESCRIPTION

Enter a description

0 / 500

SUBMIT USING THE KAGGLE API

```
> kaggle competitions submit -c 2025-ncku-ee-ml-16-classes-segmen_
```

Cancel Submit



Kaggle competition

Leaderboard

[Raw Data](#)[Refresh](#)[Public](#) [Private](#)

This leaderboard is calculated with approximately 80% of the test data. The final results will be based on the other 20%, so the final standings may be different.

#	Team	Members	Score	Entries	Last
	baseline.csv		0.515838		

Set your team name as student ID and Upload your prediction csv.



Kaggle competition

Type	Percentage	Rows
Public Leaderboard	80%	800
Private Leaderboard	20%	200
Total	100.0%	1000

The Private Leaderboard will be announced at 2025/11/26 0:00 am



Grade

- Report(50%)
- Code Implementation(30%)
- Leaderboard Ranking(20%)



Report(50%)

Data Analysis & Preprocessing (15%)

- **Dataset Exploration (5%):**
 - Describe the **dataset structure** (train/test split, number of images per class).
 - Provide **data visualization** (class distribution, example images).
- **Image Preprocessing & Augmentation (10%):**
 - Explain **how images were preprocessed** (e.g., resizing, normalization, grayscale conversion).
 - Describe any **data augmentation** techniques used to improve model performance.



Report(50%)

Model Selection & Training (20%)

- **Model Architecture(10%):**
 - Explain which **deep learning model** you chose and why (e.g. ResNet, DenseNet).
 - Describe the **architecture details** (number of layers, activation functions, optimizers).
- **Loss Function & Optimization(5%):**
 - Justify your choice of **loss function** (e.g., cross-entropy loss).
 - Explain the **optimizer** and **learning rate schedule** used.
- **Training Process & Hyperparameters(5%):**
 - Describe the **batch size, number of epochs, validation split**.
 - Discuss any **techniques used to prevent overfitting** (e.g., dropout, batch normalization).



Report(50%)

Experiment & Results (15%)

- **Comparison of Different Approaches(10%):**
 - If multiple models were tested, compare their performances.
 - Discuss **strengths and weaknesses** of your chosen model.
- **Error Analysis & Model Improvements(5%):**
 - Identify **common misclassifications** and suggest ways to improve them.
 - Discuss **possible improvements** (e.g., using pre-trained models, tuning hyperparameters).



Code Implementation(30%)

Code Quality & Documentation (15%)

- Code must be **well-structured and modular** (e.g., separate functions for preprocessing, training, evaluation).
- Proper **comments** explaining each major step.
- Use of **a clear main script or Jupyter Notebook** for execution.

Code Execution Process (15%)

- The submitted code must **include logs or screenshots of training and evaluation outputs**.
- **Charts and figures** must be generated during execution (e.g., loss curves, accuracy curves).
- The final model should **output predictions and generate a CSV file** in the required Kaggle format.



Leaderboard Ranking(20%)

Your **final submission to Kaggle** will be evaluated based on segmentation **mIoU**.
The **top-ranked models** will receive higher scores:

- No valid submission: **0 points**
- All other valid submissions: **+5 points**
- Over **public** baseline model(Unet): **+5 points**
- Top 20 of students: **+5 points**
- Top 10 of students: **+5 points**

The final ranking will be a **weighted average** of **public and private**

$$\text{Final Score} = 0.8 * \text{public} + 0.2 * \text{private}$$



Moodle Requirements

學號_姓名_HW2.zip

- report.pdf:
- code(folder): Python scripts, Jupyter Notebook, or .py files
- fig(folder): Execution Logs / Screenshots

If the format does not match, **10 points will be deducted**

Moodle Deadline: 2025/11/25 23:59:59

Kaggle Deadline: 2025/11/25 **23:59:00**

Notice: **Kaggle does not allow late submissions**